

Practice Set 1

MA628: Financial Derivatives Pricing

January 13, 2026

-
1. A 10-year German government bond (also called bund) has face value of 10,000 and an annual coupon rate of 5%. Assume that interest rate (in euros) is equal to 6% per year.
 - a. What is the bond's PV?
 - b. Suppose instead that the bund paid interest semiannually like a U.S. bond. (The bond would pay $.025 \times 10,000 = 250$ every 6 months.) What is the PV in this case?
 2. Your friend asks you to lend him \$100,000. He promises to make equal monthly payments for 10 years, 120 payments in all. You decide that a fair interest rate is 12% compounded annually. What should the monthly payment be on the \$100,000 loan?
 3. Your cousin is entering medical school next fall and asks you for financial help. He needs \$65,000 each year for the first two years. After that, he is in residency for two years and will be able to pay you back \$10,000 each year. Then he graduates and becomes a fully qualified doctor, and will be able to pay you \$40,000 each year. He promises to pay you \$40,000 for 5 years after he graduates. Are you taking a financial loss or gain by helping him out? Assume that the interest rate is 5% and that there is no risk.
 4. You are awarded \$500,000 in a lawsuit, payable immediately. The defendant makes a counteroffer of \$50,000 per year for the first three years, starting at the end of the first year, followed by \$60,000 per year for the next 10 years. Should you accept the offer if the discount rate is 12%? How about if the discount rate is 8%?
 5. You take a Rs. 100,000 loan from SBI for a period of 10 years. The annual interest rate is 8%. Find the monthly EMI that you need to pay every month to repay the loan.
 6. Retirement planning: Mr. Jones is contemplating retirement. He is 55 and his net worth now is \$2 million. He hopes that after retirement he can maintain a lifestyle that costs him \$100,000 per year in today's dollars (i.e., real dollars, inflation adjusted). If he retires, he will invest all his net worth in government bonds that yield a safe annual return of 5%. Inflation is expected to be 2% per year. Ignore taxes.
 - a. Is Mr. Jones rich enough to retire today if he lives until (i) 80 (ii) 100 (iii) 115?
 - b. Mr. Jones thinks he will live until about 100. What advice will you give him about retiring?